

A counterexample to the Besov adjoint-defect problem for almost commuting operators

Candidate solution packet for run `fa_banach_001`

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Status: candidate counterexample, likely valid. There are bounded self-adjoint almost commuting operators A, B and a real-valued $\varphi \in B_{\infty,1}^1(\mathbb{R}^2)$ for which $\varphi(A, B)^* - \varphi(A, B)$ is not trace class. This answers Problem 2 of Aleksandrov–Peller negatively.

1 Source problem

Aleksandrov and Peller define, for bounded self-adjoint A, B , the ordered double-operator-integral calculus

$$\varphi(A, B) = \iint_{\mathbb{R}^2} \varphi(x, y) dE_A(x) dE_B(y).$$

Their Problem 2, in Section 6 on PDF pages 18–19 of arXiv:1508.04702 [1], asks whether

$$(\varphi(A, B))^* - \overline{\varphi}(A, B) \in \mathcal{S}_1.$$

for every $\varphi \in B_{\infty,1}^1(\mathbb{R}^2)$, whenever A, B are bounded self-adjoint and $[A, B] \in \mathcal{S}_1$. They had proved this for the smaller projective tensor class $B_{\infty,1}^1(\mathbb{R}) \widehat{\otimes} B_{\infty,1}^1(\mathbb{R})$.

2 Idea of the proof

For two spectral values in each variable, reversing the order of the two spectral projections detects the rectangular four-point difference of φ . If both spectral gaps have size h , the commutator of the two operators costs h^2 , whereas a $B_{\infty,1}^1$ atom at that scale can have a rectangular difference of order h . One such block is harmless. The separation appears after taking many copies: at scale $h_k \asymp 2^{-k}$, use $k2^k$ identical blocks. Their commutator costs sum to a convergent series, but their adjoint defects sum to the harmonic series. A coefficient $1/k^2$ on the k -th Besov atom keeps the Besov norm summable.

3 Construction and proof

Put

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad C = [Q, P] = \frac{1}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Both P and Q are rank-one orthogonal projections, and

$$\|C\|_{\mathcal{S}_1} = 1. \tag{1}$$

For $k \geq 1$, set

$$h_k = 2^{-k-4}, \quad a_k = 4h_k, \quad N_k = k2^k,$$

and define two self-adjoint 2×2 matrices

$$A_k = (a_k - h_k/2)I + h_k P, \quad B_k = -h_k I/2 + h_k Q.$$

Their two spectral values are $a_k \pm h_k/2$ and $\pm h_k/2$, respectively, and

$$[A_k, B_k] = h_k^2 [P, Q], \quad \|[A_k, B_k]\|_{S_1} = h_k^2. \quad (2)$$

On

$$\mathcal{H} = \bigoplus_{k \geq 1} (\mathbb{C}^2)^{\oplus N_k}$$

let A and B be the direct sums of N_k copies of A_k and B_k . They are bounded and self-adjoint. By (2),

$$\|[A, B]\|_{S_1} = \sum_{k \geq 1} N_k h_k^2 = 2^{-8} \sum_{k \geq 1} k 2^{-k} = \frac{1}{128} < \infty. \quad (3)$$

Choose a real $\eta \in C_c^\infty((-3/4, 3/4))$ such that $\eta(-1/2) = 0$ and $\eta(1/2) = 1$, and put $F(u, v) = \eta(u)\eta(v)$. Define

$$\varphi(x, y) = \sum_{k \geq 1} \frac{h_k}{k^2} F\left(\frac{x - a_k}{h_k}, \frac{y}{h_k}\right). \quad (4)$$

The x -supports of the summands are disjoint: the k -th lies in $((13/4)h_k, (19/4)h_k)$, while the next lies strictly to its left.

Lemma 1. *The real function φ in (4) belongs to $B_{\infty,1}^1(\mathbb{R}^2)$.*

Proof. Let Δ_j be any fixed homogeneous Littlewood–Paley resolution used to define $B_{\infty,1}^1$. Translation does not change the norm. Since $h_k = 2^{-(k+4)}$, dyadic scaling merely shifts the Littlewood–Paley index, and therefore

$$\left\| h_k F\left(\frac{\cdot - z_k}{h_k}\right) \right\|_{B_{\infty,1}^1} \leq C_F, \quad z_k = (a_k, 0),$$

with a constant independent of k . (Equivalently, the factor h_k exactly cancels the degree-one scaling factor h_k^{-1} .) Hence the k -th summand of (4) has Besov norm at most C_F/k^2 . The series converges in $B_{\infty,1}^1$ because $\sum k^{-2} < \infty$. Its pointwise sum is the displayed continuous real function (the disjoint supports also make this identification immediate). $\square$

We next record the finite-block identity. Write $P_+ = P$, $P_- = I - P$, $Q_+ = Q$, and $Q_- = I - Q$. Given four real numbers $t_{\varepsilon\delta}$, the ordered double operator integral on one block is

$$T = \sum_{\varepsilon, \delta \in \{+, -\}} t_{\varepsilon\delta} P_\varepsilon Q_\delta.$$

Using $P_- = I - P$ and $Q_- = I - Q$ gives

$$T^* - T = (t_{++} - t_{+-} - t_{-+} + t_{--})[Q, P]. \quad (5)$$

At the four spectral pairs of (A_k, B_k) , all summands of (4) except the k -th vanish. The choice of η gives

$$\begin{array}{c|cc} & h_k/2 & -h_k/2 \\ \hline a_k + h_k/2 & h_k/k^2 & 0 \\ a_k - h_k/2 & 0 & 0 \end{array}$$

for the four values of φ . Applying (5) and (1), the trace norm of the adjoint defect on every copy of the k -th block is exactly

$$\|\varphi(A_k, B_k)^* - \varphi(A_k, B_k)\|_{\mathcal{S}_1} = \frac{h_k}{k^2}. \quad (6)$$

The common block decomposition reduces both spectral measures, so the global double operator integral is the direct sum of these block integrals. Thus

$$\begin{aligned} \|\varphi(A, B)^* - \varphi(A, B)\|_{\mathcal{S}_1} &= \sum_{k \geq 1} N_k \frac{h_k}{k^2} \\ &= \frac{1}{16} \sum_{k \geq 1} \frac{1}{k} = \infty. \end{aligned} \quad (7)$$

Since φ is real, $\overline{\varphi}(A, B) = \varphi(A, B)$. Equations (3) and (7) prove the promised counterexample.

Theorem 1. *There exist bounded self-adjoint operators A, B with $[A, B] \in \mathcal{S}_1$ and a real-valued $\varphi \in B_{\infty,1}^1(\mathbb{R}^2)$ such that*

$$\varphi(A, B)^* - \overline{\varphi}(A, B) \notin \mathcal{S}_1.$$

Consequently, Problem 2 of Aleksandrov–Peller has a negative answer.

4 Verification and novelty status

The included script `code/verify_blocks.py` checks (5), including its trace norm, on 1,000 random four-point tables and evaluates the partial sums in (3) and (7). This computation is only a regression check; the proof above is exact.

A bounded novelty search covered arXiv and exact-web queries for the paper title, “Problem 2,” the adjoint formula, “almost multiplicative,” and $B_{\infty,1}^1(\mathbb{R}^2)$, plus the citing literature visible from the journal record and Peller’s 2024 survey [2]. It found the original problem and repeated accounts of the proved commutator/trace formula, but no later proof or counterexample to this adjoint-defect question. Novelty is therefore plausible, not certified.

5 Limitations and review recommendation

This theorem completely negates Problem 2, but does not decide the separate product-defect question in Problem 1. Human review should focus first on the degree-one dyadic scaling in the Besov lemma and on the order/sign in (5). The finite-block identity is independently checked by the included script.

References

- [1] A. B. Aleksandrov and V. V. Peller, *Functions of almost commuting operators and an extension of the Helton–Howe trace formula*, J. Funct. Anal. 271 (2016), 3300–3322; arXiv:1508.04702.
- [2] V. V. Peller, *Besov spaces in operator theory*, Russian Math. Surveys 79 (2024), 1–52.

6. OPEN PROBLEMS

In this section we state two naturally arising problems.

Problem 1. It was proved in [23] that for almost commuting self-adjoint operators A and B , the functional calculus

$$\varphi \mapsto \varphi(A, B), \quad \varphi \in (L^\infty(\mathbb{R}) \hat{\otimes} B_{\infty,1}^1(\mathbb{R})) \bigcap (B_{\infty,1}^1(\mathbb{R}) \hat{\otimes} L^\infty(\mathbb{R})),$$

is *almost multiplicative*, i.e.,

$$(\varphi\psi)(A, B) - \varphi(A, B)\psi(A, B) \in \mathcal{S}_1, \quad \varphi, \psi \in (L^\infty(\mathbb{R}) \hat{\otimes} B_{\infty,1}^1(\mathbb{R})) \bigcap (B_{\infty,1}^1(\mathbb{R}) \hat{\otimes} L^\infty(\mathbb{R})).$$

Here $\hat{\otimes}$ stands for projective tensor product.

The Besov class $B_{\infty,1}^1(\mathbb{R}^2)$ does not form an algebra. However, functions $\varphi(A, B)$ of A and B depend only on the restrictions of φ to the cartesian product $\sigma(A) \times \sigma(B)$ of the spectra of A and B . It is well-known that the class of restrictions of functions in $B_{\infty,1}^1(\mathbb{R}^2)$ to a compact subset of $\mathbb{R}^2$ forms an algebra.

It would be interesting to find out whether the functional calculus $\varphi \mapsto \varphi(A, B)$, $\varphi \in B_{\infty,1}^1(\mathbb{R}^2)$, is also almost multiplicative.

Problem 2. It was proved in [23] that for almost commuting self-adjoint operators A and B , the functional calculus

$$\varphi \mapsto \varphi(A, B), \quad \varphi \in B_{\infty,1}^1(\mathbb{R}) \hat{\otimes} B_{\infty,1}^1(\mathbb{R}),$$

has the following property:

$$(\varphi(A, B))^* - \overline{\varphi}(A, B) \in \mathcal{S}_1, \quad \varphi \in B_{\infty,1}^1(\mathbb{R}) \hat{\otimes} B_{\infty,1}^1(\mathbb{R}).$$

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We would like to pose the problem to find out whether the same property holds for arbitrary functions φ in $B_{\infty,1}^1(\mathbb{R}^2)$.

Figure 1: Problem 2 in the source (PDF pages 18–19).